REGLAT at HalluScoring-2026: A Multistage Approach for Arabic Hallucination Detection

Nsrin Ashraf[1,2] and Mariam M. Abdellatif[1,2] and Mariam Labib[1,3] and Ahmed M. Fetouh[2] and Hamada Nayel[2,4]

[1]Computer Engineering, Elsewedy University of Technology, Cairo, Egypt
[2]Faculty of Computers and Artificial Intelligence, Benha University, Egypt
[3]Faculty of Engineering, Mansoura University, Egypt
[4]College of Engineering, Wadi Ad Dwaser,
Prince Sattam Bin Abdulaziz University, KSA

Abstract

This paper presents a multistage ensemble-based approach developed for the HalluScoring-2026 shared task on Arabic hallucination detection. The proposed system combines three Arabic pretrained language encoders: MARBERTv2, CAMeLBERT, and ARBERT. The diverse linguistic representations captured by these models enable the ensemble to improve the robustness and discriminative capability of hallucination detection compared to the individual encoders. On the development set, the ensemble achieved an AUC-ROC of 0.9413, Macro-F1 of 0.8704, and an AUC-PR of 0.9610, outperforming the individual models. On the official test set, the system achieved an AUC-ROC of 0.7717, Macro-F1 of 0.6576, and an AUC-PR of 0.9181. These results secured the highest overall ranking among all submitted systems, placing our system first in the HalluScoring-2026 shared task. Further analysis reveals substantial variation in detection performance across generator models and linguistic phenomena, with subtle factual deviations, dialectal and culturally specific expressions, and unusually long responses posing particular challenges. Overall, the findings demonstrate the effectiveness of combining distinct Arabic-specific pretrained encoders for robust Arabic hallucination detection.

1 Introduction

Large language models (LLMs) have achieved strong performance in question answering and text generation, but they can still produce fluent responses containing fabricated, unsupported, or factually inconsistent information, commonly known as hallucinations (Huang et al., 2025). Reliable hallucination detection is an essential task for trustworthy AI, particularly for low-resource languages such as Arabic. Compared with English, Arabic hallucination detection remains relatively underexplored due to limited benchmarks and standardized evaluation settings (Mubarak et al., 2024; Alansari and Luqman, 2025). Arabic presents additional challenges because of its rich morphology, lexical variation, and substantial differences between Modern Standard Arabic, regional dialects and domain-specific texts (Diab and Habash, 2014; Labib et al., 2026; Sobhy et al., 2025; Ehab et al., 2026). These characteristics can make subtle factual and semantic inconsistencies more difficult to identify, highlighting the need for Arabic-focused hallucination detection methods (Mubarak et al., 2024; Alansari and Luqman, 2025). The HalluScoring-2026 shared task addresses this gap by providing a standardized framework for detecting hallucinated responses in Arabic question answering.

In this work, we present a multistage ensemble approach that combines various Arabic pretrained language encoders for hallucination detection. The proposed system exploits diverse representations from multiple Arabic models through probability-level ensembling and is evaluated under the official HalluScoring setting. Our results demonstrate the effectiveness of Arabic-specific ensemble modeling for reliable hallucination detection in Arabic QA.

2 Background

The HalluScoring-2026 shared task (Alansari et al., 2026) focuses on detecting hallucinated content in Arabic responses generated by large language models (LLMs). We participate in **Task 1.1: Generalization Across Questions**, which formulates hallucination detection as a binary classification problem. Each instance contains an Arabic question q, a verified gold answer a_g, and a model-generated answer a_m. The system predicts a binary label $y \in \{0, 1\}$, where 0 denotes a non-hallucinated response and 1 denotes a hallucinated

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response. A response is non-hallucinated when it is factually consistent with the gold answer, including acceptable paraphrases or omitted details that introduce no false information. Conversely, a response is hallucinated when it contains fabricated or incorrect information, contradicts the reference, or introduces false information into an otherwise correct answer (Alansari and Luqman, 2026).

An example from the dataset illustrates the task. For the question في أي سنة تخرج نزار قباني من جامعة القاهرة, the gold answer is تخرج من جامعة دمشق, whereas the answer generated by Allam is في عام ٥٤٩١. The response is labeled 1 (hallucinated) because it provides a plausible answer that is inconsistent with the reference. This illustrates how hallucinated responses can remain linguistically fluent and plausible despite containing factual inconsistencies.

Task 1.1 specifically evaluates **generalization across questions**. Evaluation questions are completely disjoint from those in the training data, while the same generator-model architectures are represented across the splits (Alansari and Luqman, 2026). The task, therefore assesses whether systems learn transferable signals of factual consistency rather than question-specific patterns.

The HalluScoring dataset consists of Arabic question-answer instances covering diverse domains, including Islamic studies, history, geography, science, literature, and general knowledge. Each instance contains an identifier, question, gold answer, generator model, generated response, and binary hallucination label, with labels withheld for the official test set (Alansari and Luqman, 2026). In the dataset version used in our experiments, responses originate from six LLMs: Claude-Sonnet, Allam, DeepSeek-V3, Fanar, GPT-o3, and QCRI/Fanar-1-9B-Instruct. Across the combined labeled training and development data, hallucinated examples account for 53.72% of the instances. The proportion is 52.56% in the training set and 57.92% in the development set. Further details on the dataset splits are provided in Section A. Hallucination detection is important for LLM reliability because generated responses can be fluent while containing fabricated, unsupported, or factually inconsistent information (Huang et al., 2025). Despite increasing attention to hallucination detection, Arabic remains comparatively underexplored. Mubarak et al. (2024) introduced *Halwasa*, an Arabic-focused resource for analyzing hallucinations in LLM-generated text, while Alansari and Luqman (2025) proposed *AraHalluEval*, a fine-grained framework for evaluating hallucinations in Arabic generative question answering and summarization. These studies highlight the need for robust Arabic-specific hallucination detection methods. Our work addresses this need under the HalluScoring-2026 Task 1.1 setting by combining three Arabic-specific encoders—MARBERTv2, CAMeLBERT, and ARBERT—through probability-level ensembling. These models capture distinct representations of Modern Standard and dialectal Arabic (Abdul-Mageed et al., 2021; Inoue et al., 2021). By integrating their predictions, our approach aims to improve hallucination detection robustness across unseen questions and heterogeneous LLM-generated responses.

3 System Overview

Our proposed framework combines three Arabic pretrained Transformer encoders such as MARBERTv2, CAMeLBERT, and ARBERT to provide robust hallucination detection across diverse Arabic questions and LLM-generated responses. As illustrated in Figure 1, the framework follows two stages: *independent model optimization* followed by *probability-level ensemble inference*. This design preserves the specialized representations learned by each encoder while exploiting their predictions in the final decision.

In the first stage, MARBERTv2, CAMeLBERT, and ARBERT are independently fine-tuned as binary classifiers, with the best checkpoint for each model selected according to development AUC-ROC. Rather than classifying the generated response in isolation, each model jointly encodes the question Q and generated answer A_m as:

$$[\text{CLS}]; Q; [\text{SEP}]; A_m; [\text{SEP}], \quad (1)$$

allowing the classifier to assess the generated response in the context of its corresponding question. In the second stage, the three selected checkpoints independently predict hallucination probabilities for the same input. Their logits are converted to probabilities using softmax, and the hallucinated-class probabilities from MARBERTv2, CAMeLBERT, and ARBERT, denoted by $p_M(x)$, $p_C(x)$, and $p_A(x)$, respectively,

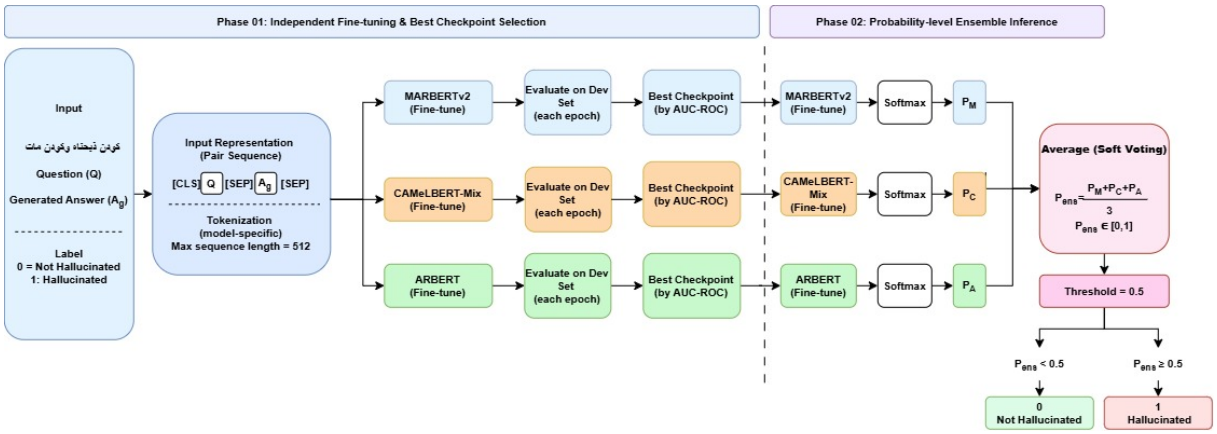


Figure 1: Overview of the proposed ensemble framework for Arabic hallucination detection.

are combined through unweighted soft voting:

$$p_{\text{ens}}(x) = \frac{p_M(x) + p_C(x) + p_A(x)}{3}. \quad (2)$$

Unlike hard voting, which discards prediction confidence by combining discrete labels, our framework directly integrates the probability estimates of the three encoders. Each model contributes equally, avoiding additional weighting parameters or ensemble-specific tuning. The final label is determined using a threshold of 0.5:

$$\hat{y}(x) = \begin{cases} 1, & p_{\text{ens}}(x) \geq 0.5, \\ 0, & p_{\text{ens}}(x) < 0.5, \end{cases} \quad (3)$$

where 1 and 0 denote hallucinated and non-hallucinated responses, respectively. We evaluate four configurations: MARBERTv2 (S1), CAMeLBERT (S2), ARBERT (S3), and the proposed ensemble (S4). S1–S3 represent the independently optimized classifiers, while S4 is the final submitted system. Importantly, the framework uses no additional task-specific labeled data beyond that provided by the shared-task organizers; external resources are limited to the three pretrained Arabic encoders and their tokenizers.

4 Experimental Setup

Our experiments are conducted on the HalluScoring-2026 Subtask 1.1 dataset (Alansari and Luqman, 2026), statistics and other details of the dataset have been reported in Appendix A. The task is formulated as a binary classification problem in which the objective is to determine whether a model-generated answer constitutes a hallucination (Label = 1) or not (Label = 0). For each instance, the classifier receives only the Arabic question and its corresponding model-generated answer as input. The verified gold answer provided in the dataset is not supplied to the model during training or inference; it is used only as part of the dataset annotation and reference framework for determining the ground-truth hallucination label.

4.1 Preprocessing and Hyperparameter Details

4.1.1 Preprocessing

We adopted a minimal preprocessing strategy to preserve the original linguistic and stylistic characteristics of the Arabic text. The HalluScoring dataset was provided with each generated answer already paired with its corresponding question; therefore, no additional question–answer construction or pairing was performed. The provided question–answer pairs were retained without aggressive normalization or linguistic transformations and were directly processed using the model-specific tokenizers available through the Hugging Face Transformers library. Specifically, CAMeLBERT[1] uses a vocabulary of 30,000 tokens, while MARBERTv2[2] and ARBERT[3] each use vocabularies of 100,000 tokens. For all three models, the maximum input sequence length was set to 256 tokens, with sequences exceeding this limit truncated during tokenization. This minimal preprocessing setup was intentionally maintained to preserve potentially informative lexical and stylistic cues relevant to hallucination detection.

[1] https://huggingface.co/CAMeL-Lab/bert-base-arabic-camelbert-mix
[2] https://huggingface.co/UBC-NLP/MARBERTv2
[3] https://huggingface.co/UBC-NLP/ARBERT

4.1.2 Hyperparameter Configuration

All three BERT-based models were fine-tuned using the same hyperparameter configuration to ensure a consistent comparison. Appendix B summarizes the training setup. Focal Loss (Lin et al., 2017) with $\gamma = 2.0$ was employed primarily to emphasize difficult and ambiguous examples. In preliminary development experiments, Focal Loss slightly improved ensemble AUC-ROC compared with standard cross-entropy (0.9413 vs. 0.9397), and was therefore retained in the final configuration. Final predictions are obtained using the unweighted soft-voting ensemble described in Section 3, with a decision threshold of 0.5. During development, we additionally evaluated Qwen2.5-1.5B-Instruct with LoRA as an exploratory baseline, and a detailed description of this baseline is given in Appendix C.

Experiments were conducted on Google Colab using an NVIDIA Tesla T4 GPU (15.6 GB VRAM). The implementation uses PyTorch (Paszke et al., 2019) and Hugging Face Transformers (Wolf et al., 2020), with a fixed random seed of 42 for reproducibility.

4.2 Task Evaluation Metrics Summary

Following the HalluScoring-2026 evaluation protocol, AUC-ROC is used as the primary ranking metric, while Macro-F1 provides a class-balanced evaluation. We additionally report AUC-PR and per-class precision, recall, and F1-score. We assess pairwise differences between model predictions on the development set using McNemar's test at a significance level of $\alpha = 0.05$. Class 0 represents *Not Hallucinated*, while class 1 represents *Hallucinated*.

5 Results

In this section, we present and interpret the experimental results of our Arabic hallucination detection system, discussing findings alongside the results rather than as a separate block.

We report the performance of the four system configurations: MARBERTv2, CAMeLBERT, ARBERT, and their soft-voting ensemble together with an exploratory LoRA-tuned Qwen2.5-1.5B-Instruct development baseline on the HalluScoring-2026 Subtask 1 dataset, using AUC-ROC as the primary metric alongside Macro-F1 and AUC-PR. We then compare the individual configurations and examine the factors behind their relative performance, including the ensemble's gains from diversity, variance reduction, and calibration. We next examine how detection difficulty varies across the six generator models represented in the data, before turning to a detailed error analysis of misclassification patterns, probability distributions, and linguistic factors.

5.1 Comparative Performance Across System Configurations

Table 1 compares the performance of the three Arabic-specific encoders, the proposed soft-voting ensemble, and the exploratory Qwen2.5-1.5B-Instruct baseline. The three Arabic-specific encoders achieve comparable development performance, with AUC-ROC scores above 0.92, while the proposed ensemble (S4) obtains the strongest development results. Based on this performance, the ensemble was selected as the final system for official shared-task submission and subsequently **ranked first** on the HalluScoring-2026 test set. The LoRA-adapted Qwen2.5-1.5B-Instruct model was evaluated only as an exploratory development baseline using the provided training and development splits and was not included in the final shared-task submission. Its lower development performance compared with the Arabic-specific encoders and their ensemble further supported the selection of the ensemble as the submitted system.

Table 1: Performance comparison of individual models and the ensemble configuration.

Configuration	Model	AUC-ROC	Macro-F1	AUC-PR
S0	Qwen2.5-1.5 + LoRA	0.9121	0.8244	0.9337
S1	MARBERTv2	0.9258	0.8551	0.9495
S2	CAMeLBERT	0.9274	0.8649	0.9434
S3	ARBERT	0.9268	0.8629	0.9453
S4	**Ensemble-Dev**	**0.9413**	**0.8704**	**0.9610**
S4	**Ensemble-Test**	**0.7717**	**0.6576**	**0.9181**

To determine whether the observed performance differences between the evaluated classifiers were statistically significant, we conducted pairwise McNemar tests on the development-set predictions using a significance level of $\alpha = 0.05$. Here, n_{01} denotes the number of instances for which the first model in the comparison was incorrectly classified while the second model was correctly classified, whereas n_{10} denotes the number of instances for which the first model was correct and the second was incorrect. The complete pairwise statistical comparison and discussion is given in Appendix F.

5.2 Model and Generator-Specific Performance Analysis

To examine whether the difficulty of hallucination detection varies across generator models, we analyze MARBERTv2 performance on responses from the six generators. As shown in Table E, performance varies substantially, with AUC-ROC ranging from 0.8707 to 0.9928. MARBERTv2 performs best on Claude-Sonnet responses (AUC-ROC = 0.9928; Macro-F1 = 0.9599), while QCRI/Fanar-1-9B-Instruct is the most challenging generator (AUC-ROC = 0.8707; Macro-F1 = 0.7884). The remaining generators achieve AUC-ROC scores between 0.9674 and 0.9784. The generator-specific class distributions also vary considerably, indicating that the observed performance differences may reflect both generator-specific response characteristics and differences in class balance and sample size. Overall, the results highlight substantial generator-dependent variation that may be obscured by aggregate performance metrics. Appendix D summarizes the predicted hallucination probabilities in detail.

6 Conclusion

This paper presented REGLAT, an ensemble of MARBERTv2, CAMeLBERT, and ARBERT for Arabic hallucination detection in HalluScoring-2026. The ensemble achieved an AUC-ROC of 0.9413, Macro-F1 of 0.8704, and AUC-PR of 0.9610 on the development set and ranked **first** in the official shared-task evaluation. Our analysis further showed that detection difficulty varies across generator models and linguistic characteristics, particularly for subtle factual errors, dialectal expressions, and extreme response lengths. These findings demonstrate the value of combining three Arabic-specific encoders for robust hallucination detection. Future work will explore dialect-aware modeling and improved detection of subtle factual inconsistencies.

7 limitations

To better understand the limitations of the proposed system, we analyzed misclassified development instances and predicted probability distributions. False positives often involve factually correct responses with uncommon information, elaborate paraphrases, or non-standard linguistic forms. Dialectal and culturally specific expressions also present challenges; for example, instances 507 and 508 contain the colloquial expression كودن ذبحناه وكودن مات. False negatives typically occur when otherwise plausible responses contain subtle factual errors, such as incorrect dates, numerical values, or attributions, or when incorrect information is embedded within an otherwise correct answer. Some errors may also reflect differences in reference-answer specificity. For example, one development instance asked ما الحدث الذي يُعد نهاية رمزية للحرب الباردة؟. The gold answer was قمة مالطا (١٩٨٩), whereas the generated response stated سقوط جدار برلين عام ١٩٨٩ يعتبر حدثًا رمزيًا لنهاية الحرب الباردة. هذا الحدث أدى إلى إعادة توحيد ألمانيا وساهم بشكل كبير في انهيار الاتحاد السوفيتي وتفككه لاحقاً، مما وضع حداً رسمياً للتوترات والصراع بين القوى العظمى خلال تلك الفترة. Although the generated response mentions the historically correct year 1989, it identifies the fall of the Berlin Wall rather than the Malta Summit as the reference event and was therefore labeled as hallucinated. The model nevertheless assigned a hallucination probability of only 0.0003, illustrating the difficulty of detecting subtle event-level factual mismatches in otherwise fluent and plausible responses.

The analysis further reveals systematic effects related to linguistic variation and response characteristics. Questions containing Gulf or Egyptian Arabic expressions produce higher error rates for MARBERTv2 and ARBERT, whereas CAMeLBERT is more robust on these cases. Mid-length responses (approximately 30–50 words) are classified more accurately than very short (< 10 words) or very long (> 80 words) responses, while responses containing explicit uncertainty markers are more frequently classified as hallucinated. These observations suggest that the models rely on both semantic information and linguistic or stylistic cues, including dialect, response length, and expressions of uncertainty.

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Appendices

A Data Split Usage

The dataset is organized into two predefined splits: a training set comprising 4,705 samples and a development set containing 1,300 samples. The training set was used for model fine-tuning, while the development set was employed for model evaluation and selection. The detailed class distribution and statistics of both splits are summarized in Table 2. The dataset exhibits a relatively balanced class distribution, with a slightly higher proportion of Hallucinated instances. No separate labeled test set was provided; therefore, the development set was used to evaluate and compare the proposed models while the test set was kept blind.

Split	Total	Halluc.	Not Halluc.
Train	4,705	2,473 (52.56%)	2,232 (47.44%)
Dev	1,300	753 (57.92%)	547 (42.08%)
Total	**6,005**	**3,226 (53.72%)**	**2,779 (46.28%)**

Table 2: Class distribution across the training and development sets.

B Unified hyperparameter configuration used for all BERT-based models

Hyperparameter	Value
Batch Size	16
Learning Rate	2×10^{-5}
Max Epochs	5
Early Stopping Patience	3
Max Sequence Length	256
Optimizer	AdamW
Scheduler	Cosine with Warmup
Warmup Ratio	0.1
Weight Decay	0.01
Loss Function	Focal Loss ($\gamma = 2.0$)
Mixed Precision	FP16
Random Seed	42

C Exploratory Qwen2.5 Baseline

Qwen2.5-1.5B-Instruct with LoRA has been impleented to assess whether parameter-efficient adaptation of a generative LLM could provide competitive performance on the task. This experiment was conducted only on the provided training and development splits and was not part of the final submitted system. The model was trained for three epochs with a maximum sequence length of 384, learning rate of 1×10^{-4}, training batch size of 1, evaluation batch size of 4, and 16 gradient accumulation steps. LoRA was configured with rank $r = 32$, $\alpha = 64$, and dropout of 0.05, with a warmup ratio of 0.1, weight decay of 0.01, and random seed of 42.

D Predicted Hallucination Probabilities

Table 3 summarizes the predicted hallucination probabilities. The individual encoders produce mean probabilities between 0.5138 and 0.5214 and cover nearly the full probability range. The ensemble has a slightly lower standard deviation (0.3812) than the individual models (0.3821–0.3895), indicating a modest smoothing effect from probability averaging. Overall, the most difficult cases involve subtle factual deviations, dialectal and culturally specific expressions, extreme response lengths, and differences in reference-answer specificity.

Table 3: Probability distribution characteristics of the individual models and ensemble.

Model	Min Probability	Max Probability	Mean	Std Dev
CAMeLBERT	0.0001	0.9999	0.5214	0.3821
MARBERTv2	0.0016	0.9972	0.5138	0.3895
ARBERT	0.0015	0.9982	0.5182	0.3858
Ensemble	0.0023	0.9989	0.5178	0.3812

E Per-generator class distribution and MARBERTv2 performance on the development set

Generator	N	Halluc.	Not Halluc.	AUC-ROC	Macro-F1	AUC-PR
Claude-Sonnet	100	47 (47%)	53 (53%)	0.9928	0.9599	0.9930
Allam	100	78 (78%)	22 (22%)	0.9784	0.8645	0.9937
DeepSeek-V3	100	67 (67%)	33 (33%)	0.9769	0.9430	0.9906
Fanar	100	85 (85%)	15 (15%)	0.9710	0.9169	0.9946
GPT-o3	100	92 (92%)	8 (8%)	0.9674	0.8464	0.9971
QCRI/Fanar-1-9B-Instruct	800	384 (48%)	416 (52%)	0.8707	0.7884	0.8719
Overall	**1,300**	**753 (57.92%)**	**547 (42.08%)**	**0.9258**	**0.8551**	**0.9495**

F Pairwise McNemar Test

Table 4 reports results of the pairwise McNemar test on the development set. The McNemar analysis shows that the ensemble differs significantly from MARBERTv2 ($\chi^2 = 7.446$, $p = 0.0064$). In contrast, the differences between the ensemble and CAMeLBERT ($p = 0.1307$) and between the ensemble and ARBERT ($p = 0.0637$) are not statistically significant at the 0.05 level. Pairwise differences among the individual encoders are also non-significant. These results indicate that although the ensemble achieves the strongest overall development metrics, its improvement is statistically clear relative to MARBERTv2, while the gains over CAMeLBERT and ARBERT are more modest.

Table 4: Pairwise McNemar test results on the development set.

Model Comparison	n_{01}	n_{10}	χ^2	p-value
MARBERTv2 vs. Ensemble	44	21	7.446	0.0064
MARBERTv2 vs. CAMeLBERT	66	57	0.520	0.4707
MARBERTv2 vs. ARBERT	64	56	0.408	0.5228
Ensemble vs. CAMeLBERT	30	44	2.284	0.1307
Ensemble vs. ARBERT	21	36	3.439	0.0637
CAMeLBERT vs. ARBERT	65	66	0.000	1.0000

A noticeable performance gap is observed between the development and official test sets. The proposed ensemble decreases from an AUC-ROC of 0.9413 and Macro-F1 of 0.8704 on the development set to 0.7717 and 0.6576, respectively, on the hidden test set. Importantly,

this degradation was not unique to REGLAT. The official leaderboard shows that all participating systems experienced a decrease in AUC-ROC from development to test evaluation, with drops ranging from approximately 13.4 to 21.7 percentage points. This consistent pattern suggests that the hidden test set represented a substantially more challenging generalization setting for participating systems. Despite this shift, REGLAT achieved the highest official test AUC-ROC and secured first place in the shared task, indicating comparatively strong generalization under the more difficult hidden-test conditions.